

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
First Semester Master of Technology (CE)
University Examination, May 2011
CE707: Distributed Computing

Date: 13/05/2011

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 70

Instructions:

1. Figure to the right indicates full marks
2. Make suitable assumption whenever necessary and mention it clearly
3. Parts of same question should be answered together and in the same sequence.
4. Answers to the sections must be written separately.

SECTION-I

Q-1 Answer the followings:

- [A] What is distributed system? Give at least one example of each types of distributed system. List types of computer used in distributed system. [03]
- [B] What is significance of state management in web development? List and explain any two methods for state management. [02]
- [C] What is XML? How it is differ from HTML? Give example of XML code. [02]
- [D] Define SOAP. Explain SOAP model with diagram. [04]

Q-2 Attempt any two:

1. Explain following paradigms of distributed computing with respect to architecture and technique / method: [12]
- The Peer-to-Peer Paradigm
 - The Message Passing Paradigm
 - The Point-To-Point Message Model of message system paradigm
- Give example of each paradigm.
2. .Net component can be called from a COM component. Write steps to do so. Each step should include syntax and example.
3. Explain layout of a SOAP response with source code format. Briefly describe each element of it.

Q-3 Attempt the following questions:

- [A] What is Cloud computing? Explain Cloud Computing architecture and any one industry frameworks. [03]
- [B] What is Grid computing? Explain Gird Architecture with diagram. [03]
- [C] Write steps to create and consume XML web service in .Net with example. [06]

OR

- [C] Explain concept of ActiveX DLL, write a code for ActiveX DLL which takes input as any one arithmetic operator and two operands and return result of this operation. Write steps and code for an application which use this ActiveX DLL. [06]

SECTION-II

Q-4 Attempt following:

1. Write a java client server echo application using Stream socket. [04]
2. Describe java RMI architecture. What is role of RMI registry? [04]
3. Explain needs of data representation in IPC. [03]

Q-5 Attempt following:

- [A] Write Java IDL to build server/client suite that implement day-time protocol. [06]
- [B] Explain key methods and constructor of Socket class. What is use of SocketImpl in stream programming? [06]

OR

- [B] Write RMI client steps and program to download file from server. [06]

Q-6 Attempt any two:

1. What is web service? What is purpose of UDDI in web services architecture? Explain UDDI architecture in detail [12]
2. Explain HTTP protocol events and HTTP message header with diagram. Justify statement "HTTP is a connection-oriented and Stateless protocol"
3. What is ORB? Draw and describe architecture of CORBA.